MOSS CORE V1C

60 x 65 mm / four corner fasteners / USB rover controller

PROTOTYPE — ERC/DRC do not validate analogue, firmware or real rover behavior.

01 USB Power

USB-C device. Current-limited input + reverse blocking + 3.3V buck. RF disabled. Encoder supply OFF before enumeration and in suspend. USB power limits require firmware and bench verification.

File: 01_USB_Power.kicad_sch

02 Controller

ESP32-S3-WROOM-1U-N8 only. Dedicated latch feedback, USB current fault and switched encoder power. USB ROM programming via BOOT/RESET; debug TX only. Firmware port required.

File: 02_Controller.kicad_sch

03 Hardware Interlock

Hardware clear on watchdog, NC loop, external fault or supervised reset. Read back BOTH latches. Hardware-fault response must never use the host auto-rearm error code. Explicit recovery required.

File: 03_Hardware_Interlock.kicad_sch

04 Drive Encoders

MDD10A command connector order is Core-specific. Cross-map harness to Cytron labels. Encoder A and B both inverted: phase relation retained. Motor power bypasses Core.

File: 04_Drive_Encoders.kicad_sch

05 Servo

3.3V TTL TX/RX reproduces the installed Waveshare Bus Servo Adapter (A) interface. Ioff buffers. Servo power is external. Release TX after UART TX_DONE; discard echo.

File: 05_Servo.kicad_sch

06 Audio

External MAX98357A. No direct 5V-to-MCU sense divider. Hardware output-enable inhibits I2S when amplifier supply is absent. SD_MODE current limiting. Host audio compatibility requires explicit firmware work.

File: 06_Audio.kicad_sch

07 Current Sense

3S pack only. External 2mOhm Kelvin shunt >=3W subject to derating. Fuse thin Kelvin wires AT SOURCE. INA226 measures total pack current, not individual motors.

File: 07_Current_Sense.kicad_sch

Power flags identify the passive-fed rails only; they are not protection devices.







MOSS / Laurent & Codex

Sheet: /

File: MOSS_Core_V1C.kicad_sch

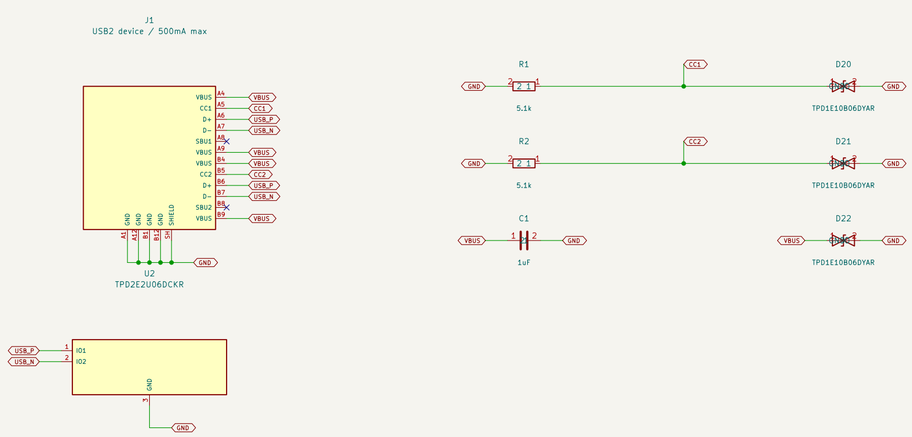
Title: MOSS Core V1C / system overview

Size: A3	Date: 2026-10-15	**Rev: V1C**
KiCad E.D.A. 10.0.6		Id: 1/8

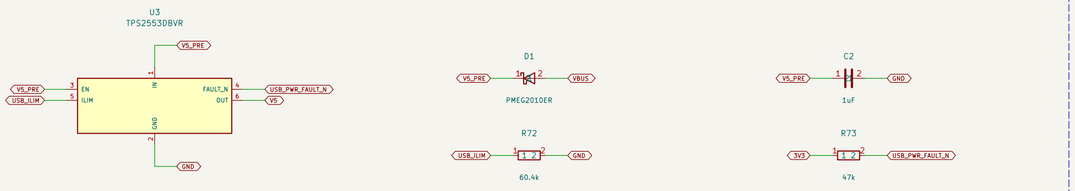
01 USB Power

USB-C device. Current-limited input + reverse blocking + 3.3V buck. RF disabled. Encoder supply OFF before enumeration and in suspend. USB power limits require firmware and bench verification.

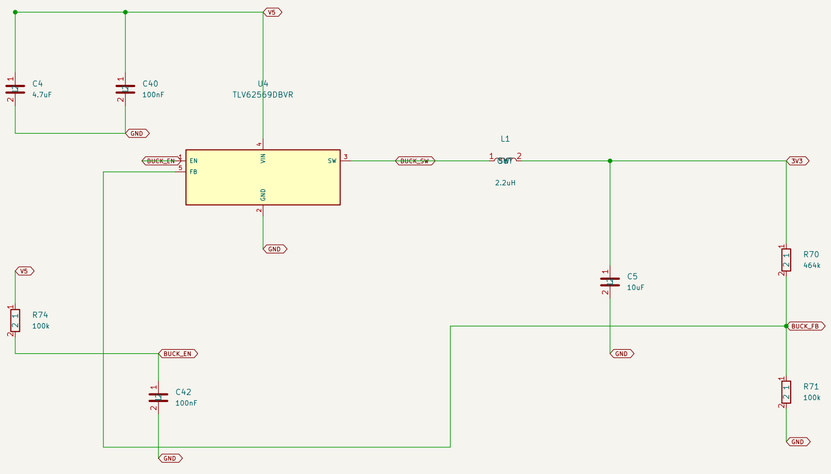
USB-C, CC and ESD



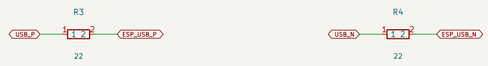
Host reverse blocking and current limit



3.3 V buck / enable / feedback



USB series resistors, at MCU



MOSS / Laurent & Codex

Sheet: /01 USB Power/
File: 01_USB_Power.kicad_sch

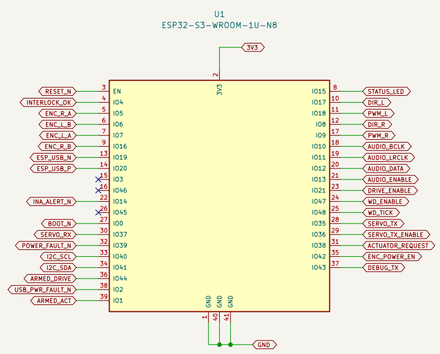
Title: MOSS Core V1C / 01 USB Power

Size: A2	Date: 2026-10-15	**Rev: V1C**
KiCad E.D.A. 10.0.6		Id: 2/8

02 Controller

ESP32-S3-WROOM-1U-N8 only. Dedicated latch feedback, USB current fault and switched encoder power, USB ROM programming via BOOT/RESET; debug TX only. Firmware port required.

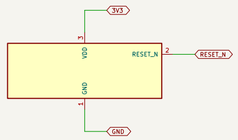
ESP32-S3 module / local supply







Supervised reset



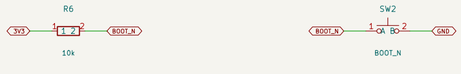






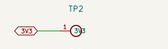
BOOT and status LED





I2C pull-ups and service pads





















Mechanical interface: M2.5, D8 reservation







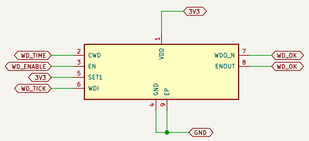


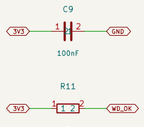
MOSS / Laurent & Codex	
Sheet: /02 Controller/	
File: 02_Controller.kicad_sch	
Title: MOSS Core V1C / 02 Controller	
Size: A2	Date: 2026-10-15
KiCad E.D.A. 10.0.6	Id: 3/8
	Rev: V1C

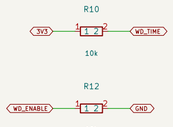
03 Hardware Interlock

Hardware clear on watchdog, NC loop, external fault or supervised reset. Read back BOTH latches. Hardware-fault response must never use the host auto-rearm error code. Explicit recovery required.

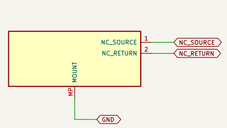
External watchdog / 200 ms nominal

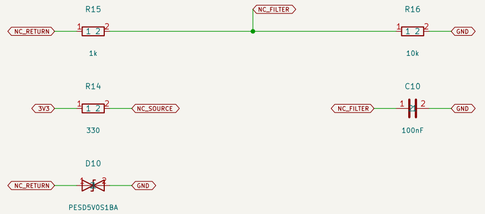




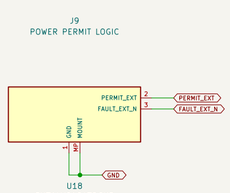


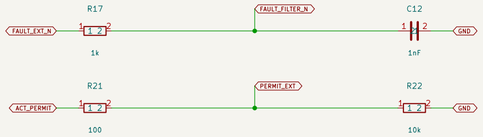
Normally closed contact input





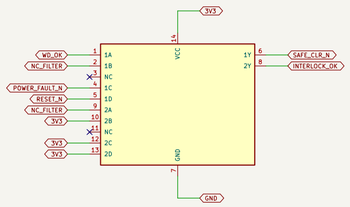
External power permission and fault receiver





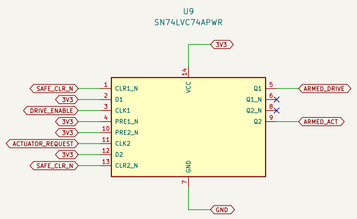


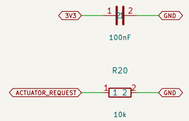
Healthy gate: WD / NC / fault / reset

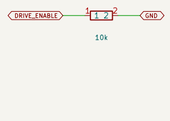




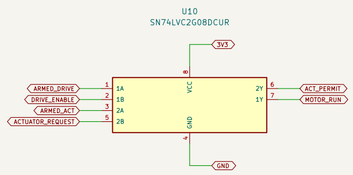
Fresh-edge arming latches

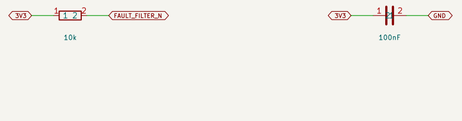






Requested and latched permissions





MOSS / Laurent & Codex
Sheet: /03 Hardware Interlock/
File: 03_Hardware_Interlock.kicad_sch

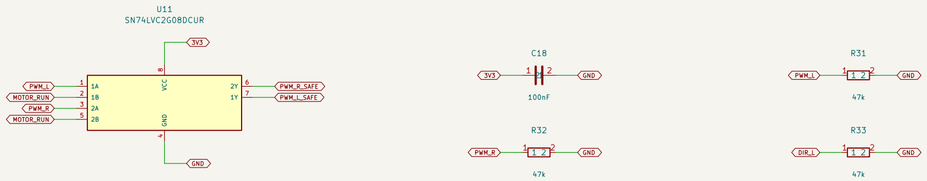
Title: MOSS Core V1C / 03 Hardware Interlock

Size: A2	Date: 2026-10-15	**Rev: V1C**
KiCad E.D.A. 10.0.6		Id: 4/8

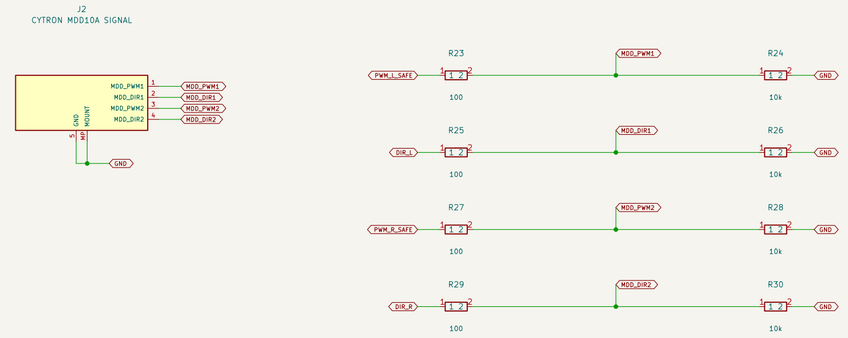
04 Drive Encoders

MDD10A command connector order is Core-specific. Cross-map harness to Cytron labels. Encoder A and B both inverted: phase relation retained. Motor power bypasses Core.

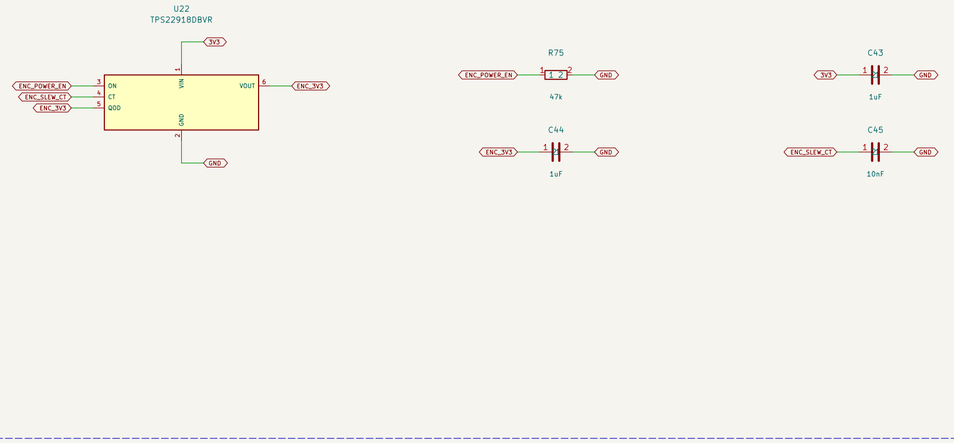
Motor request defaults and PWM mask



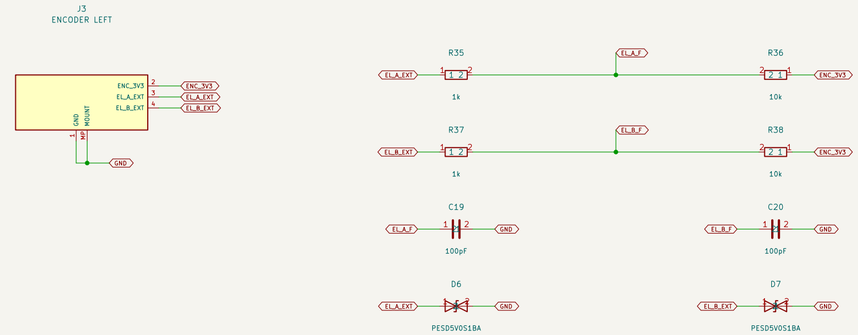
Cytron logic outputs: no motor power



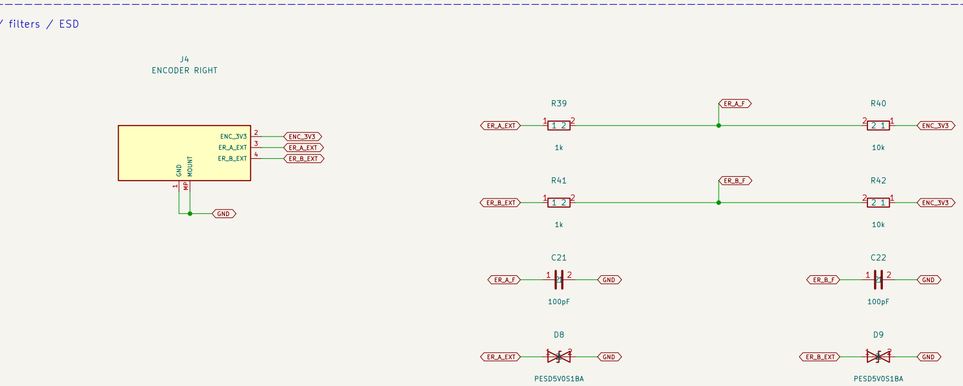
Encoder supply OFF until USB configured



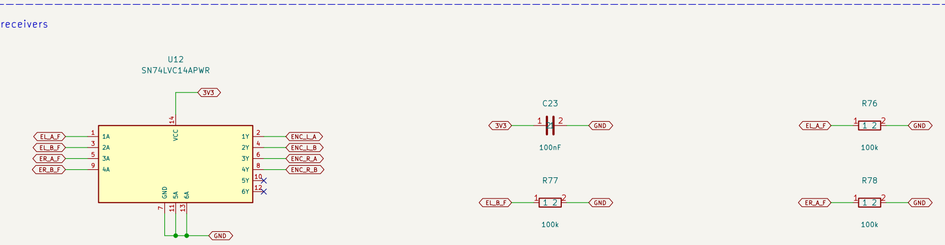
Left encoder / filters / ESD



Right encoder / filters / ESD



Encoder Schmitt receivers



MOSS / Laurent & Codex
Sheet: /04 Drive Encoders/
File: 04_Drive_Encoders.kicad_sch

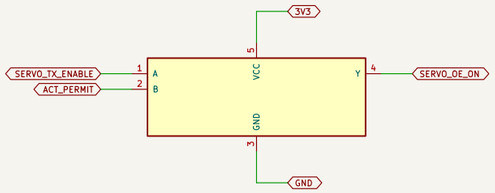
Title: MOSS Core V1C / 04 Drive Encoders

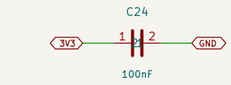
Size: A2	Date: 2026-10-15	Rev: V1C
KiCad E.D.A. 10.0.6		Id: 5/8

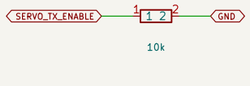
05 Servo

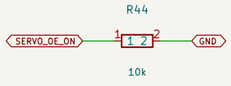
3.3V TTL TX/RX reproduces the installed Waveshare Bus Servo Adapter (A) interface. Ioff buffers. Servo power is external. Release TX after UART TX_DONE; discard echo.

Hardware TX permission

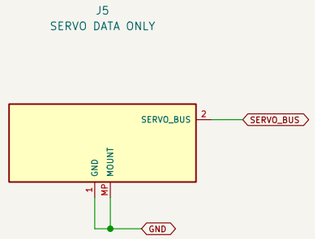


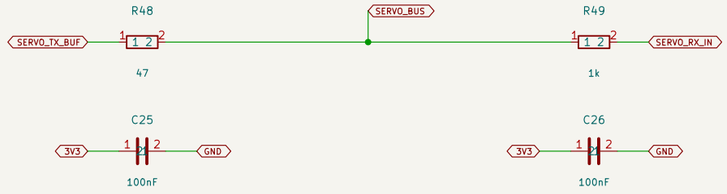


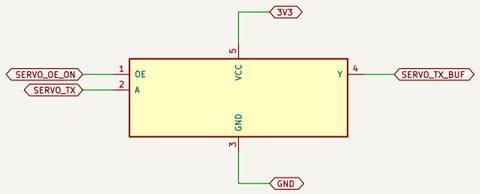


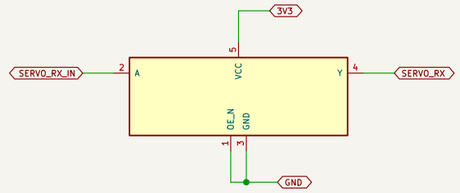


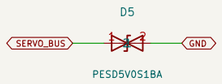
3.3 V half-duplex: data and GND only

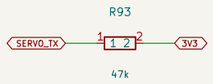












MOSS / Laurent & Codex

Sheet: /05 Servo/
File: 05_Servo.kicad_sch

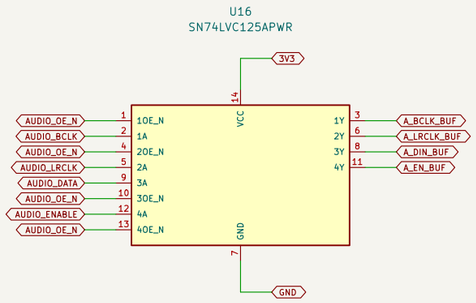
Title: MOSS Core V1C / 05 Servo

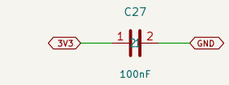
Size: A3	Date: 2026-10-15	**Rev: V1C**
KiCad E.D.A. 10.0.6		Id: 6/8

06 Audio

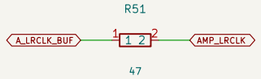
External MAX98357A. No direct 5V-to-MCU sense divider. Hardware output-enable inhibits I2S when amplifier supply is absent. SD_MODE current limiting. Host audio compatibility requires explicit firmware work.

I2S outputs / defaults / series resistors

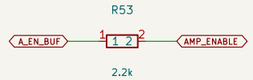




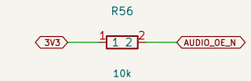


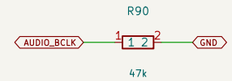


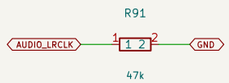


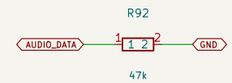




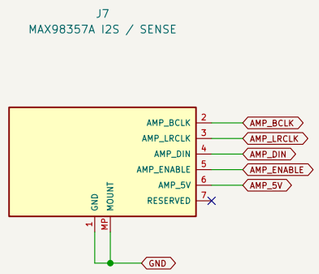


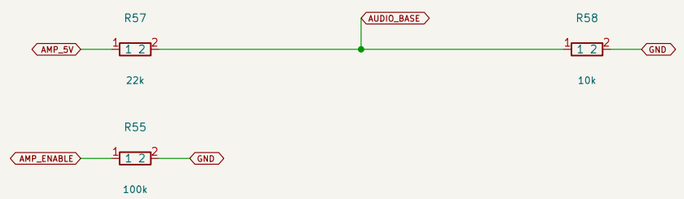


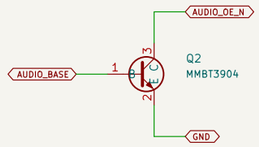




External amplifier and supply detection







MOSS / Laurent & Codex

Sheet: /06 Audio/
File: 06_Audio.kicad_sch

Title: MOSS Core V1C / 06 Audio

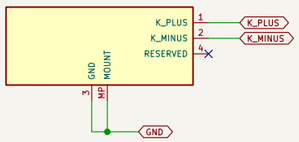
Size: A3	Date: 2026-10-15	**Rev: V1C**
KiCad E.D.A. 10.0.6		Id: 7/8

07 Current Sense

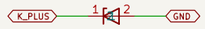
3S pack only. External 2mOhm Kelvin shunt >=3W subject to derating. Fuse thin Kelvin wires AT SOURCE. INA226 measures total pack current, not individual motors.

Kelvin entry / source protection external

J6
KELVIN 3S BATTERY



D3
SMF15A

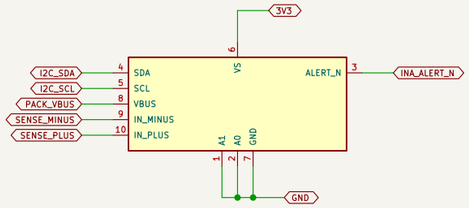


D4
SMF15A

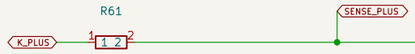


INA226 / input filters / I2C

U17
INA226AIDGSR



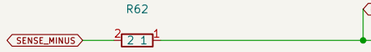
R61
10



C30
100nF



R62
10



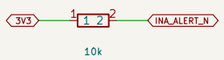
R63
10



C29
100nF

C31
100nF

R64
10k



MOSS / Laurent & Codex
Sheet: /07 Current Sense/
File: 07_Current_Sense.kicad_sch
Title: MOSS Core V1C / 07 Current Sense
Size: A3 | Date: 2026-10-15 | **Rev: V1C**
KiCad E.D.A. 10.0.6 | Id: 8/8